

HILO INTL, HI, USA

WMO: 912850

Lat: 19.719N

Lon: 155.053W

Elev: 38

StdP: 14.68

Time Zone: -10.00 (W10)

Period: 94-19

WBAN: 21504

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	61.7	62.9	51.4	56.3	65.0	53.4	60.8	66.1	23.8	73.4	20.3	73.9	6.1	220	0.278

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	12.8	86.3	74.6	85.2	74.0	84.2	73.6	77.1	82.9	76.1	82.1	75.3	81.3	10.3	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
75.3	132.7	80.0	74.2	128.1	79.2	73.3	124.0	78.5	40.4	83.0	39.4	81.9	38.7	81.6	82.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
16.7	14.8	12.7	DB	58.9	88.3	1.4	2.1	57.8	89.8	57.0	91.0	56.2	92.2	55.2	93.7	
			WB	53.0	77.9	1.4	1.9	51.9	79.3	51.1	80.3	50.3	81.4	49.3	82.8	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	74.2	71.6	71.5	72.1	72.7	74.2	75.3	76.4	76.8	76.7	75.9	74.1	72.4	
	DBStd	2.77	2.02	2.31	2.19	2.05	1.84	1.56	1.79	1.75	2.08	1.91	2.06	1.91	
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0	
	CDD50	8817	671	602	684	681	750	759	819	831	800	803	723	694	
	CDD65	3342	206	182	219	231	285	309	354	366	350	338	273	229	
	CDH74	17422	819	693	818	905	1460	1762	2274	2430	2325	1924	1195	817	
	CDH80	2796	78	69	74	67	180	252	441	510	541	376	151	57	
Wind	WSAvg	6.4	6.6	7.0	6.9	6.7	6.5	6.3	6.3	6.2	6.2	6.1	6.1	6.2	
Precipitation	PrecAvg	125.5	9.2	10.4	13.1	12.2	8.5	6.8	9.4	10.5	8.7	10.0	14.7	12.0	
	PrecMax	211.3	32.2	45.6	49.9	43.2	25.0	22.7	28.6	48.9	25.0	23.0	45.9	35.2	
	PrecMin	62.3	0.1	0.5	0.9	1.0	0.9	1.8	3.5	2.7	1.6	2.4	1.0	0.3	
	PrecStd	29.8	8.0	9.7	10.5	8.5	5.4	3.6	5.0	7.4	4.9	4.8	9.7	7.9	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	84.4	84.8	84.3	83.4	85.1	85.3	86.4	87.2	88.2	86.8	85.5	83.5	
		MCWB	70.6	70.9	71.9	71.7	72.5	73.0	74.9	76.0	76.3	74.8	73.9	72.1	
	2%	DB	82.1	82.1	82.1	81.8	83.4	83.8	85.0	85.5	86.2	85.2	83.3	81.6	
		MCWB	70.3	70.6	71.0	71.1	72.2	72.9	74.2	74.9	74.9	74.3	73.5	71.4	
	5%	DB	80.3	80.2	80.3	80.5	82.0	82.6	83.8	84.3	84.7	83.7	81.8	80.1	
		MCWB	69.7	69.8	70.2	70.5	71.7	72.5	73.8	74.3	74.2	73.8	72.7	70.5	
	10%	DB	78.6	78.4	78.6	79.1	80.7	81.5	82.7	83.1	83.3	82.2	80.2	78.6	
		MCWB	69.0	69.0	69.4	69.9	71.1	72.0	73.3	73.7	73.6	73.2	71.8	69.8	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	73.3	73.4	73.6	74.2	75.2	75.5	77.5	78.3	78.3	77.4	76.1	74.4
			MCDB	80.0	80.0	80.5	80.2	81.6	81.9	83.1	84.0	84.8	83.4	82.2	80.3
2%		WB	72.1	72.3	72.4	72.6	73.6	74.5	76.2	76.8	76.9	76.0	74.8	73.1	
		MCDB	78.7	78.6	79.1	78.6	80.3	81.0	82.0	82.2	83.2	82.2	80.7	78.6	
5%		WB	71.0	71.2	71.4	71.6	72.8	73.5	75.3	75.8	75.7	75.0	73.7	72.0	
		MCDB	77.6	77.4	77.9	77.8	79.5	80.0	81.1	81.4	81.9	81.2	79.3	77.3	
10%		WB	70.1	70.2	70.4	70.8	71.9	72.9	74.4	74.9	74.7	74.0	72.9	71.1	
		MCDB	76.4	76.2	76.5	76.9	78.5	79.3	80.3	80.6	80.9	80.0	78.1	76.2	
Mean Daily Temperature Range	5% DB	MDBR	14.5	14.0	13.2	12.7	13.4	13.1	13.1	12.8	13.7	13.3	12.7	13.1	
		MCDBR	17.4	16.9	15.9	14.7	15.0	14.2	14.2	14.3	15.1	14.9	14.8	15.2	
		MCWBR	11.1	10.4	9.6	8.0	8.2	7.6	7.4	8.0	8.7	8.9	9.3	9.3	
	5% WB	MCDWR	15.2	14.9	14.4	13.3	13.7	13.1	12.8	12.6	13.6	13.5	13.0	13.2	
		MCWBR	10.4	9.7	9.3	7.7	7.8	7.4	7.3	7.7	8.3	8.5	8.5	8.9	
Clear-Sky Solar Irradiance	taub	0.369	0.371	0.389	0.433	0.408	0.396	0.403	0.388	0.385	0.382	0.379	0.362		
	taud	2.558	2.553	2.498	2.332	2.446	2.471	2.456	2.513	2.517	2.554	2.564	2.600		
	Ebn at Noon	289	294	292	278	282	282	280	287	288	287	284	288		
	Edh at Noon	30	32	35	42	37	36	36	34	34	31	30	28		
All-Sky Solar Radiation	RadAvg	1423	1549	1701	1880	1979	2029	2060	2001	1868	1627	1395	1290		
	RadStd	85	102	183	128	121	74	88	94	76	64	77	121		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (5 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+225	+225

Nomenclature: See separate page